



Alumni: Do Tran Nhat Tuong



2018-2022: Automotive Engineering
Đại học Sư phạm Kỹ thuật TP HCM, Việt Nam



2022-2023: AIVIETNAM Course
AIVIETNAM, Việt Nam



2023-Present: Computer Science
National Yang Ming Chiao Tung University, Taiwan

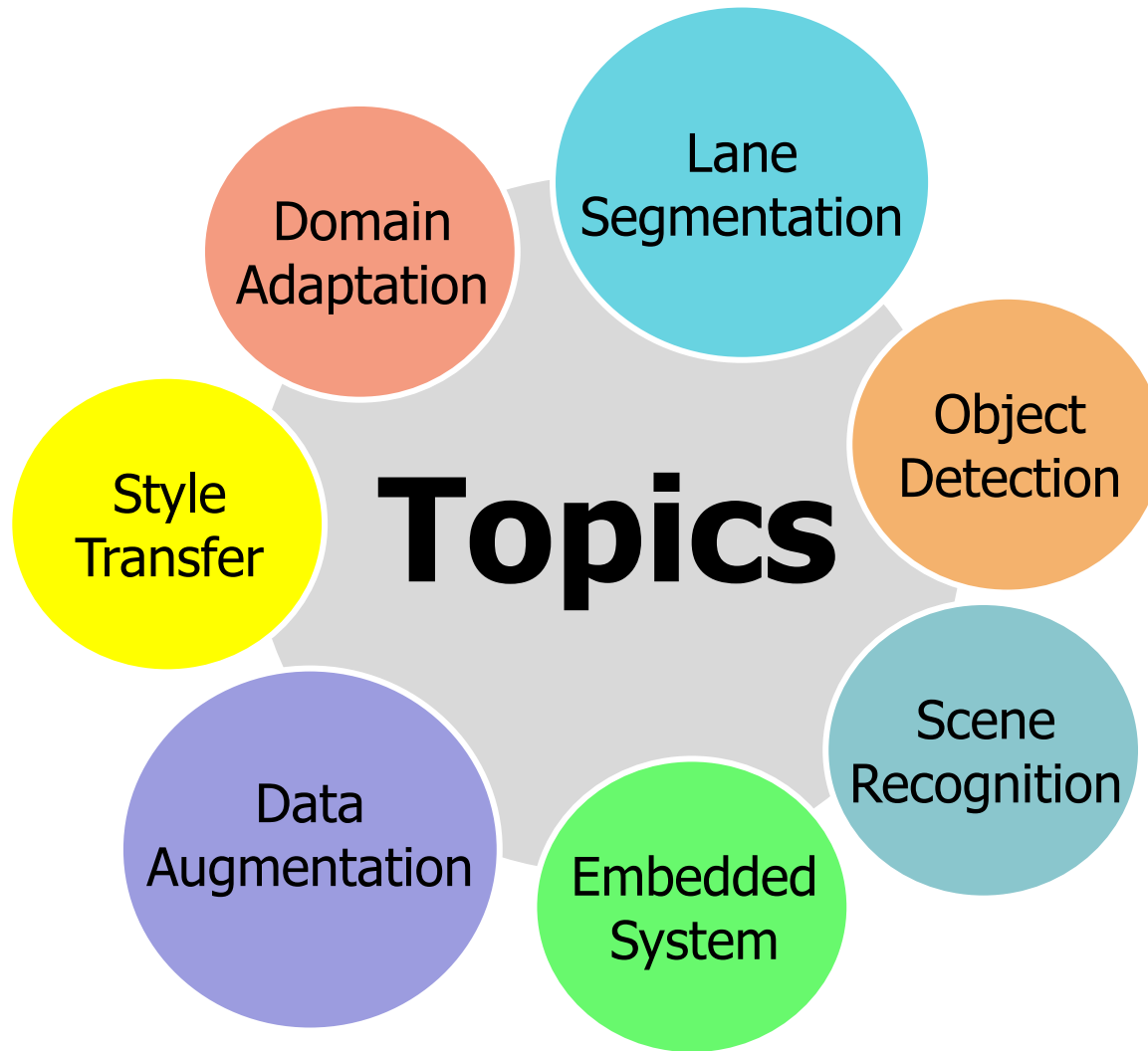
Advisor:

1. Hae-Gon Jeon: <https://scholar.google.com/citations?user=Ei00xroAAAAJ&hl=en>.
2. Tae Jong Choi: <https://dblp.org/pid/150/1657.html>.
3. Ching-Chun Huang: <https://scholar.google.com/citations?user=xTdexhsAAAAJ&hl=zh-TW>.
4. Ba Hung Ngo: https://scholar.google.com.tw/citations?user=_9iFmuYAAAAJ&hl=vi&oi=sra.
5. Vinh Quang Dinh: <https://dblp.org/pid/142/0382.html>.
6. Vu-Hoang Tran: <https://scholar.google.com/citations?user=qJ7mWjoAAAAJ&hl=en>.

Facebook: <https://www.facebook.com/tuong.do.735>

Google Scholar: <https://scholar.google.com/citations?user=UUT7AIUAAAAJ>

1. Research Direction



2. Experiences

◆ Challenges

1. 09/2021: UIT Self-Driving Car Contest – Won the second prize.
2. 02/2022: UTE Self-Driving Car Contest – Won the second prize.
3. 07/2022: Smart Menu Quy Nhon AI Challenge.
4. 09/2022: UIT Self-Driving Car Contest – Won the first prize.
5. 10/2022: AI Challenge HCMC – Won the third prize.
6. 02/2023: AI NVIDIA Challenge Track 2 – Won the third prize.
7. 04/2023: Bosch Future Mobility Challenge.

◆ Publications

1. **Nhat-Tuong Do-Tran**, Vu-Hoang Tran, Tuan-Ngoc Nguyen and Thanh-Le Nguyen. "Efficient video retrieval method based on transition detection and video metadata information." in **ICSSE**, 2023
2. Bach Hoang Ngo, Dat Thanh Nguyen, **Nhat-Tuong Do-Tran**, Phuc Pham Huy Thien, Minh-Hung An, Tuan-Ngoc Nguyen, Loi Nguyen Hoang, Vinh Dinh Nguyen, Vinh Dinh. "Comprehensive Visual Features and Pseudo Labeling for Robust Natural Language-Based Vehicle Retrieval." in Proceedings of the IEEE/CVF **CVPR Workshops**, 2023.
3. Huu-Phu Do, Yan-An Chen, **Nhat-Tuong Do-Tran**, Kai-Lung Hua, Wen-Hsiao Peng, Ching-Chun Huang. "Indirect: invertible and discrete noisy image rescaling with enhancement from case-dependent textures." **Multimedia Systems**, 30(2), 75, 2024.
4. Ba Hung Ngo, **Nhat-Tuong Do-Tran**, Tuan-Ngoc Nguyen, Hae-Gon Jeon, Tae Jong Choi. "Learning CNN on ViT: A Hybrid Model to Explicitly Class-specific Boundaries for Domain Adaptation." in Proceedings of the IEEE/CVF **CVPR**, 2024.

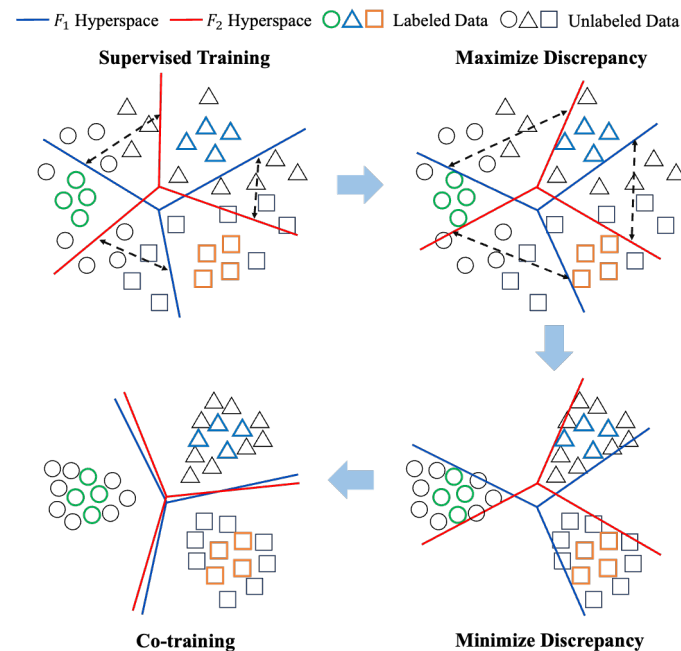
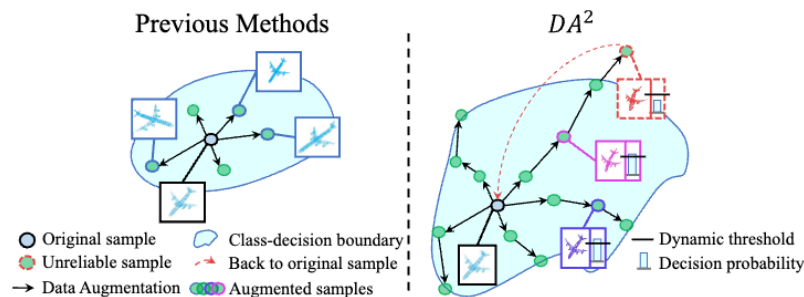
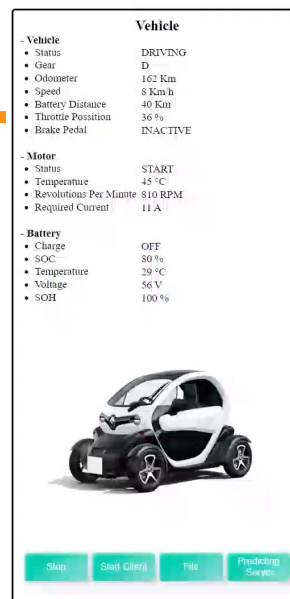
2. Experiences

◆ 2021-2023: Applications

1. Autonomous
2. Smart Menu
3. Retrieval System

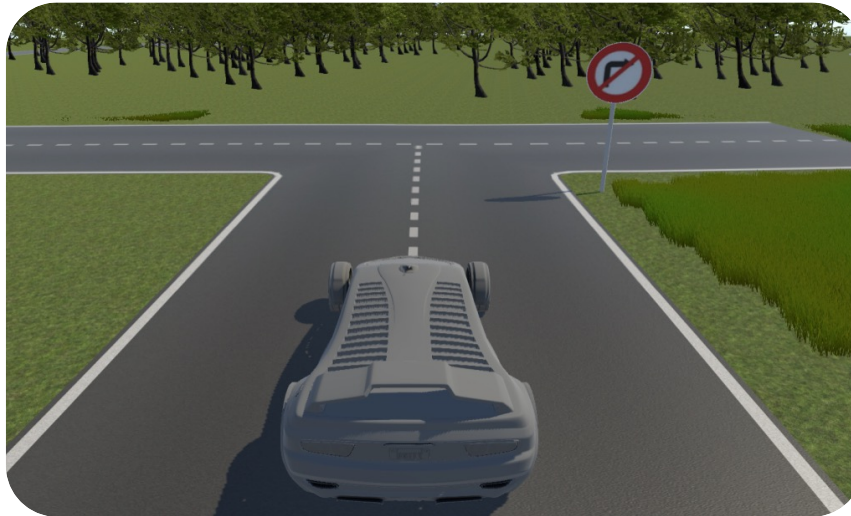
◆ 2023-2024: Research

1. Domain Adaptation
2. Data Augmentation
3. Style Transfer



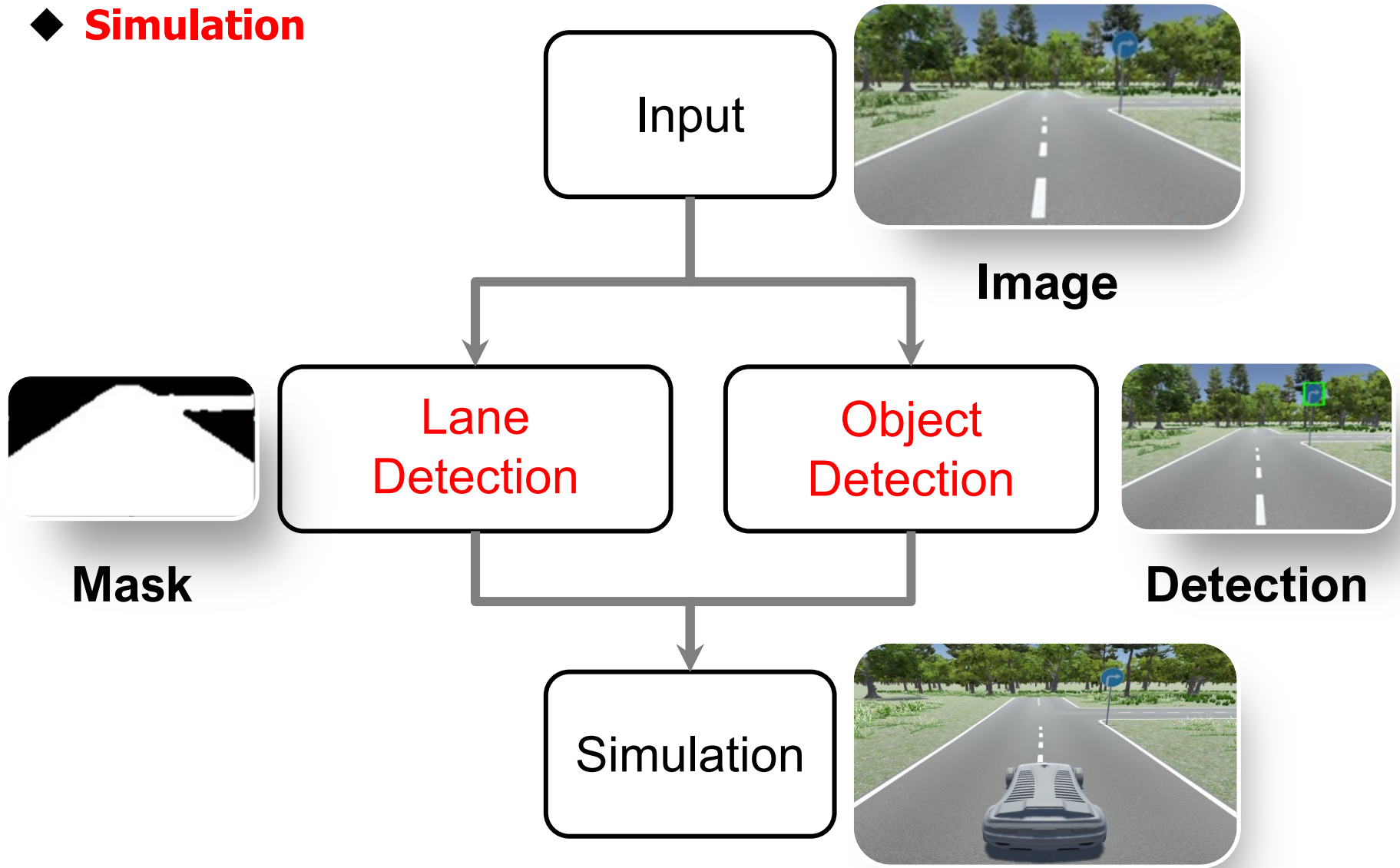
3. Autonomous

◆ Simulation



3. Autonomous

◆ Simulation



3. Autonomous

◆ Simulation

Lane Detection

Experiment Results

Mô hình	IoU(%)	FPS (RTX 3090)
ENet	90.2	98.26
Bisenet	92.1	105.80
UNet	94.8	49.63

3. Autonomous

◆ Simulation

Lane Detection

Visualization

Image



Bisenet



Unet



3. Autonomous

◆ Simulation

Object Detection

REALTIME

Experiment Results

Model	Size	Precision	Recall	mAP@0.5	mAP@0.5:0.95	FPS (RTX 3090)
YOLOv5m	640	0.995	0.997	0.993	0.861	256.41
YOLOv6n	640	0.994	0.99	0.996	0.855	307.69
YOLOv7	640	0.995	0.997	0.993	0.884	243.90

3. Autonomous

◆ Simulation

Object Detection

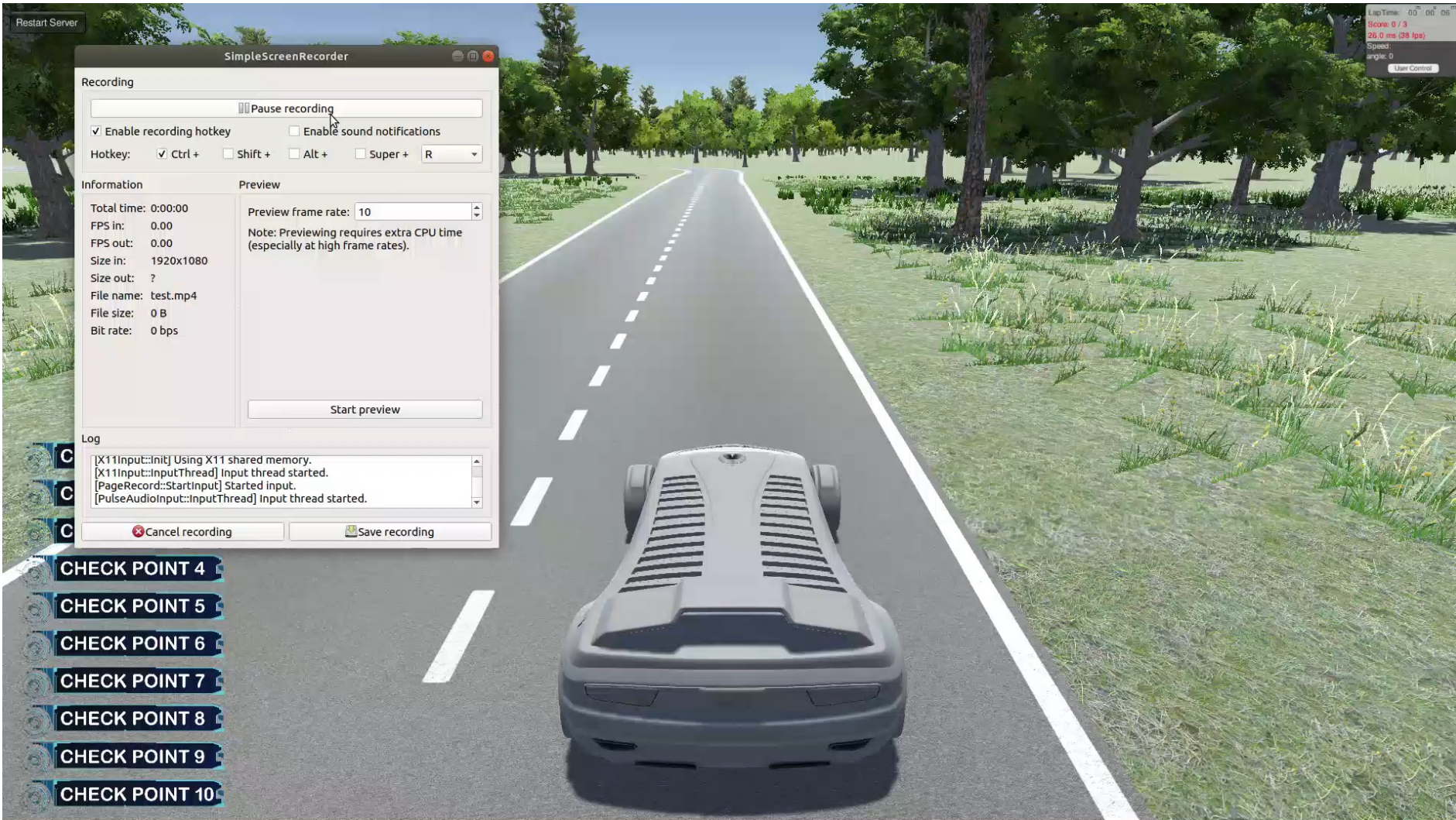


Visualization



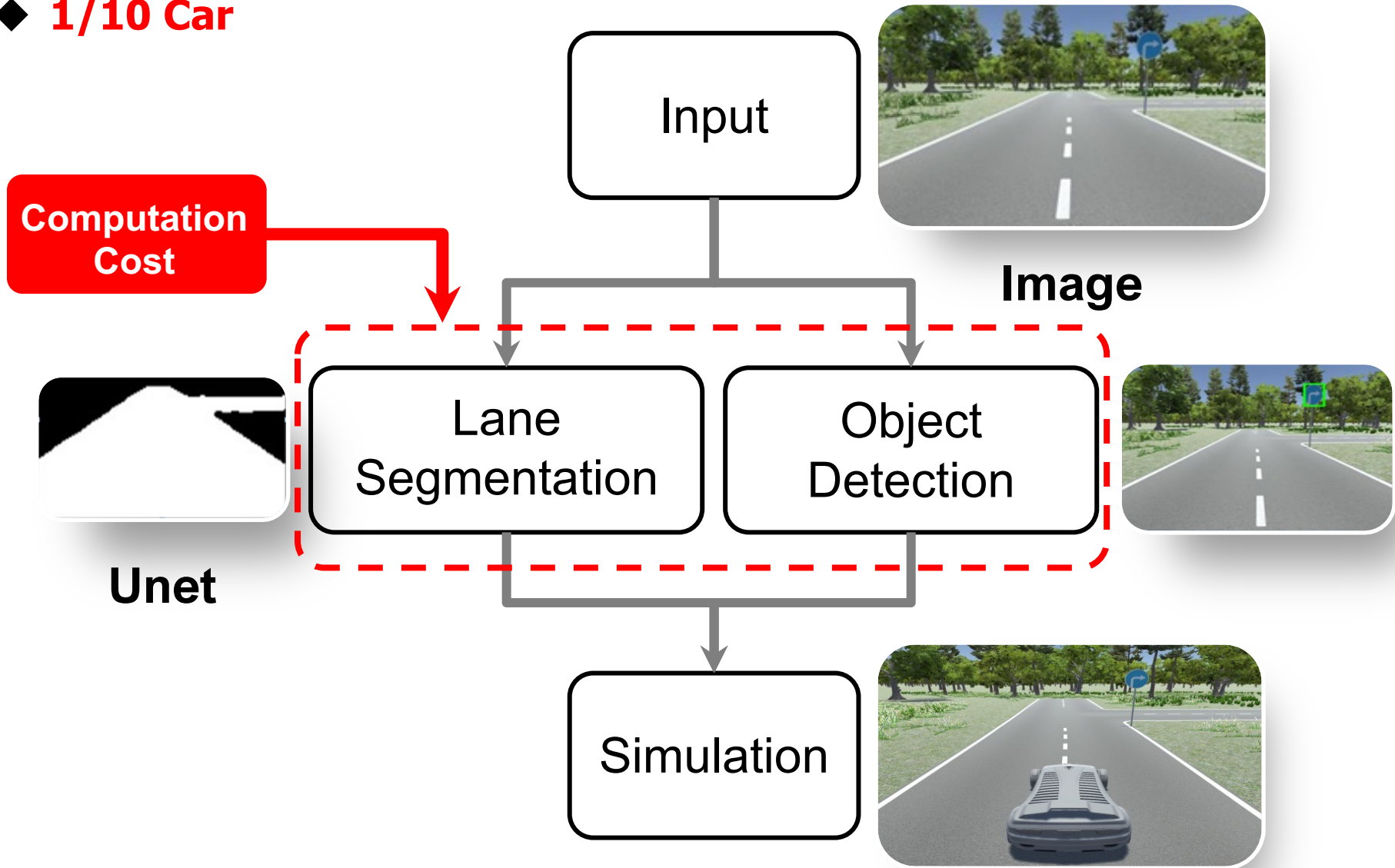
3. Autonomous

◆ Simulation



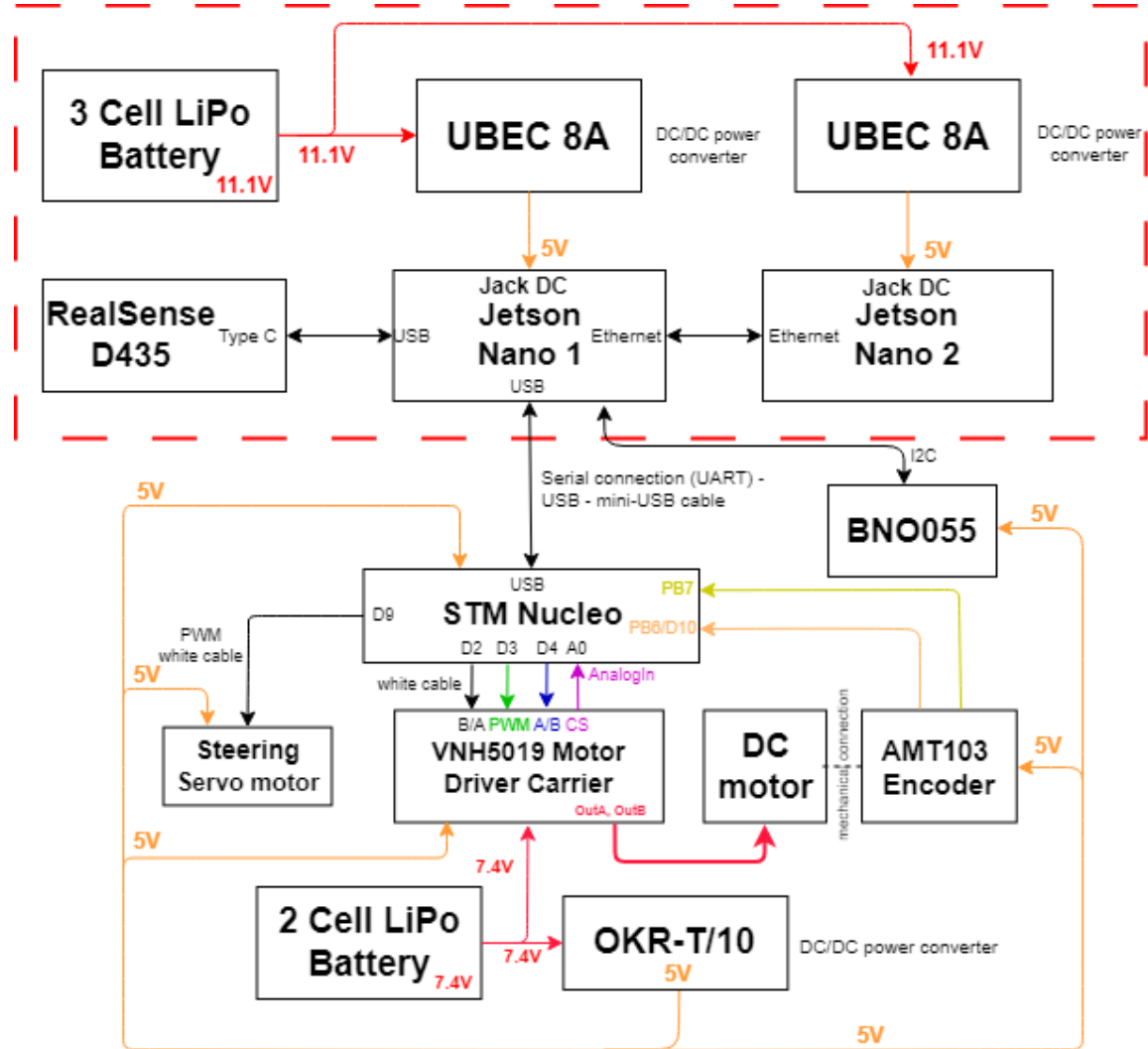
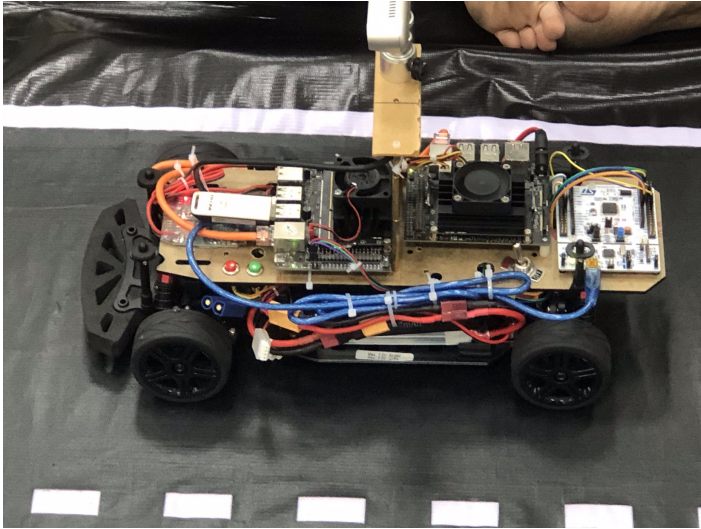
3. Autonomous

◆ 1/10 Car



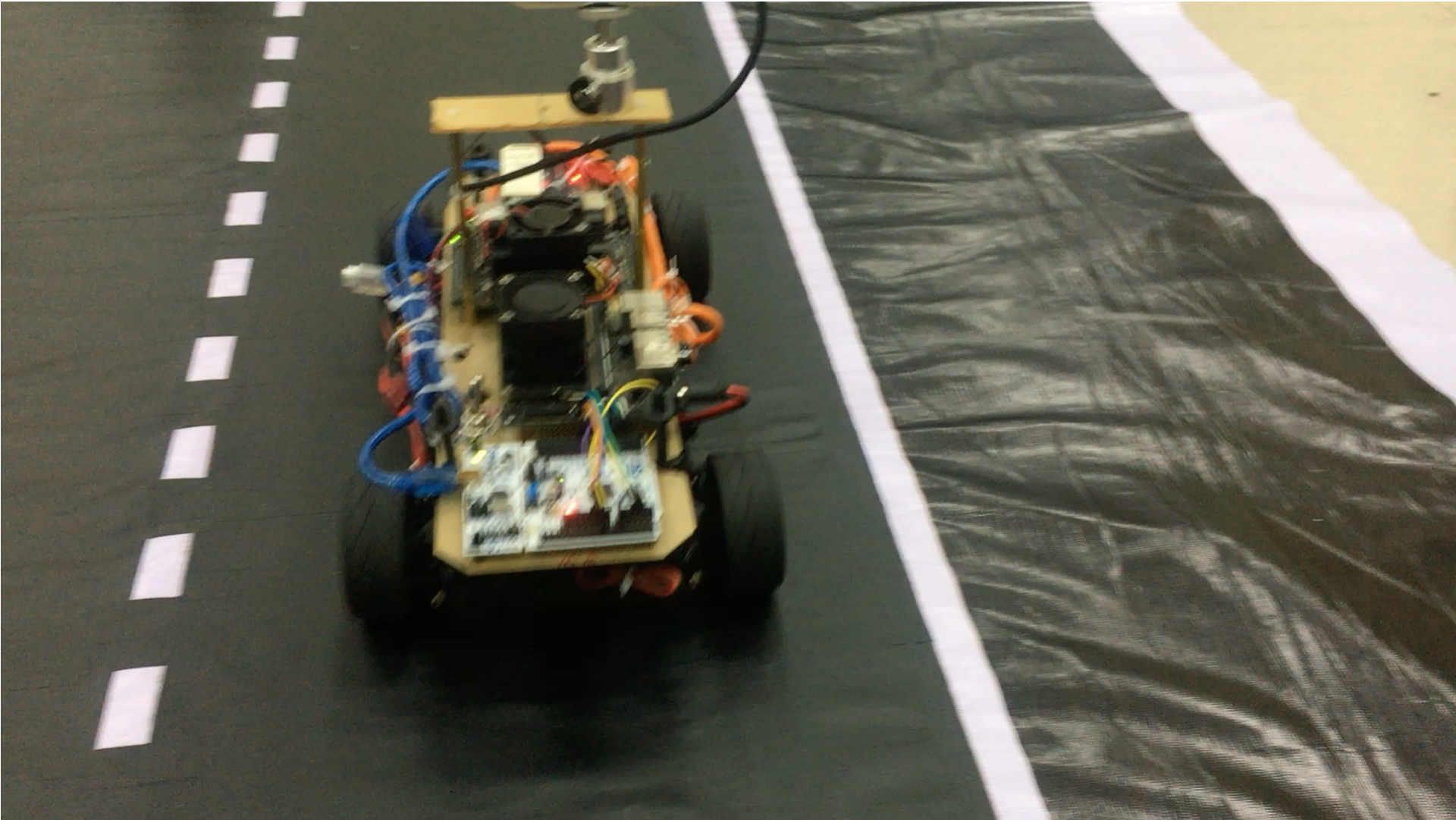
3. Autonomous

◆ 1/10 Car



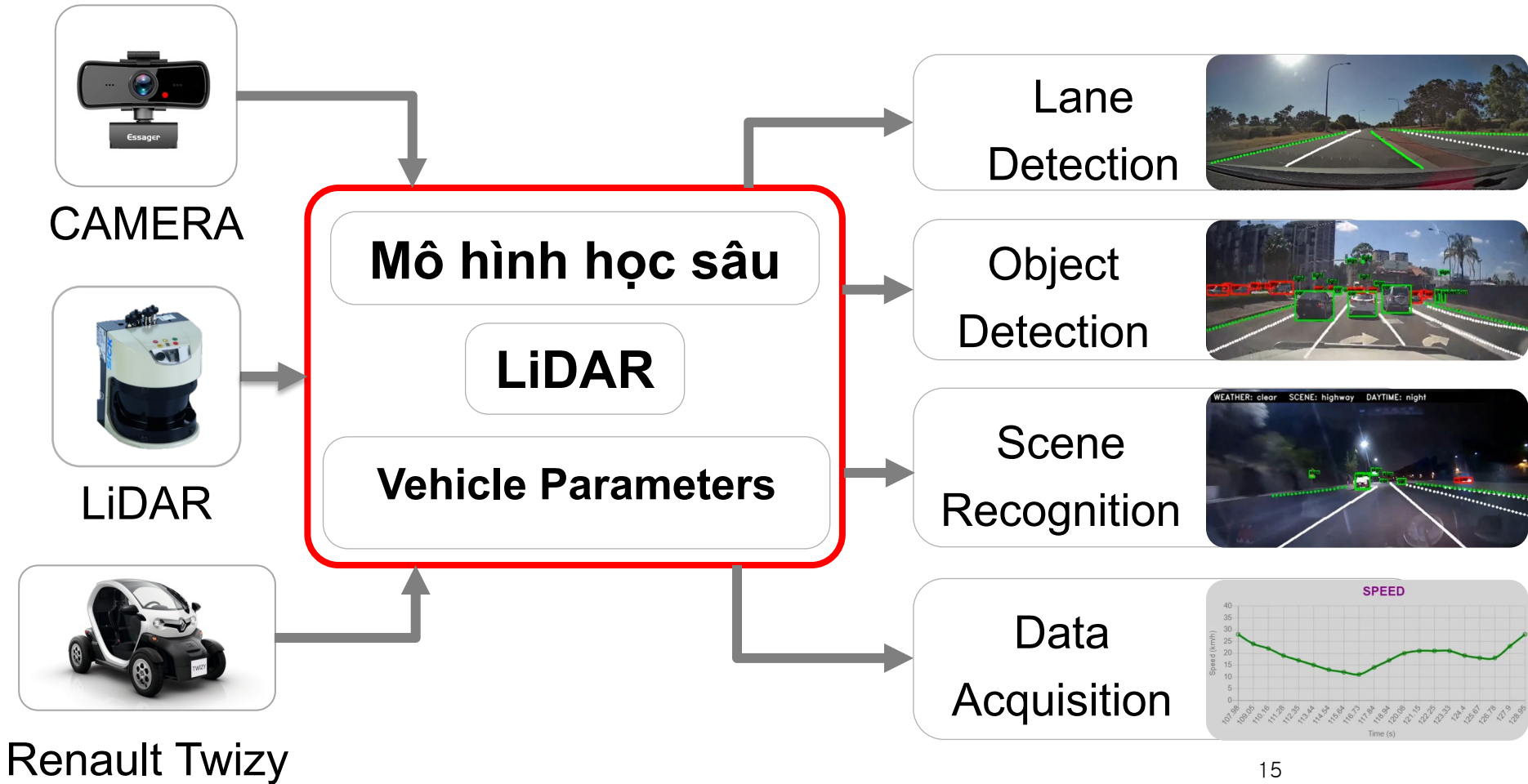
3. Autonomous

◆ 1/10 Car



3. Autonomous

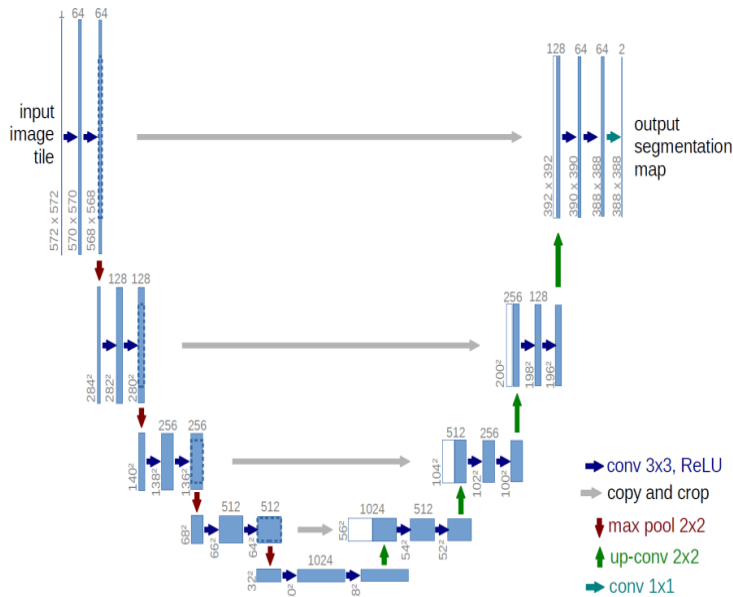
◆ Real-World



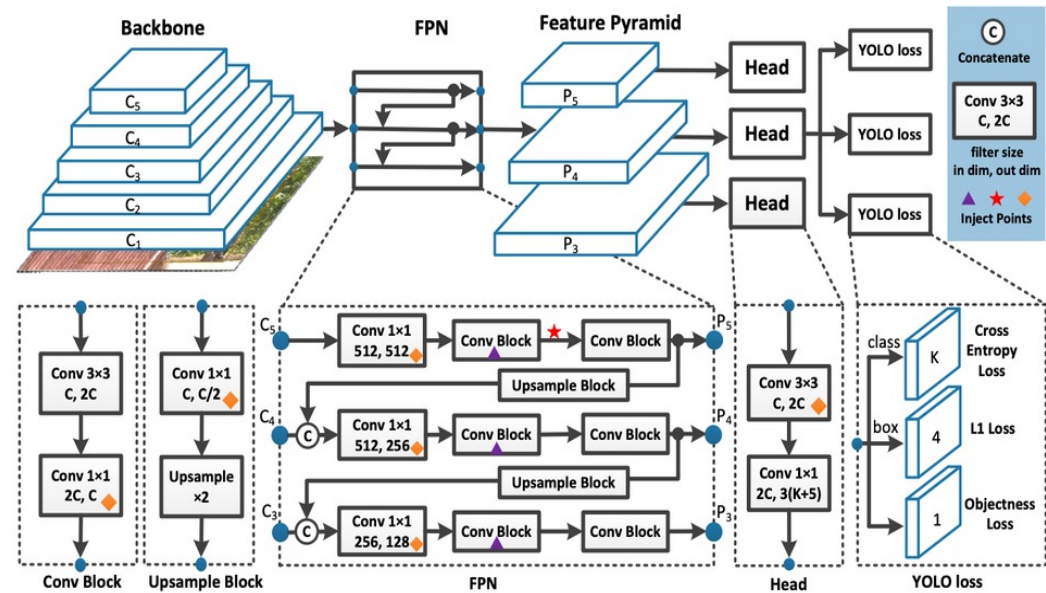
3. Autonomous

◆ Real-World

Problems



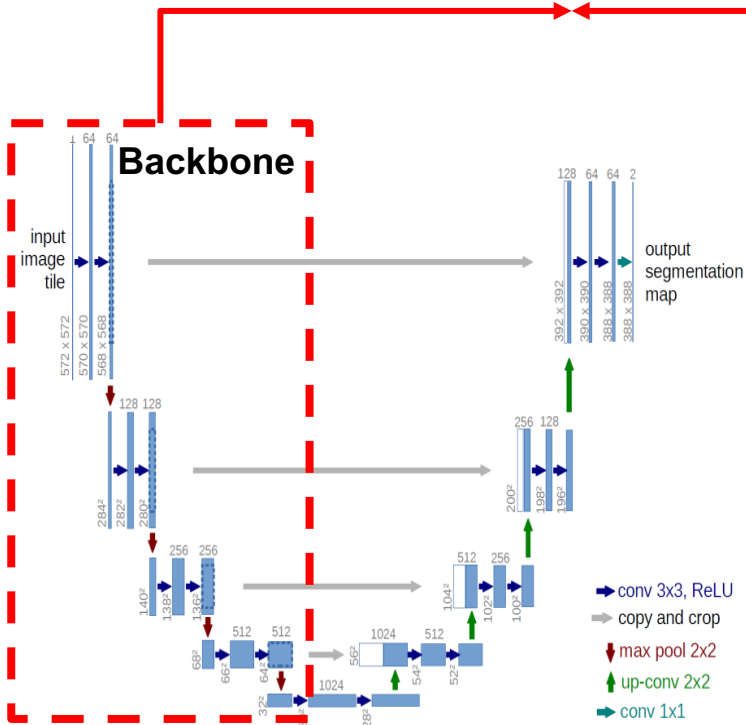
Unet



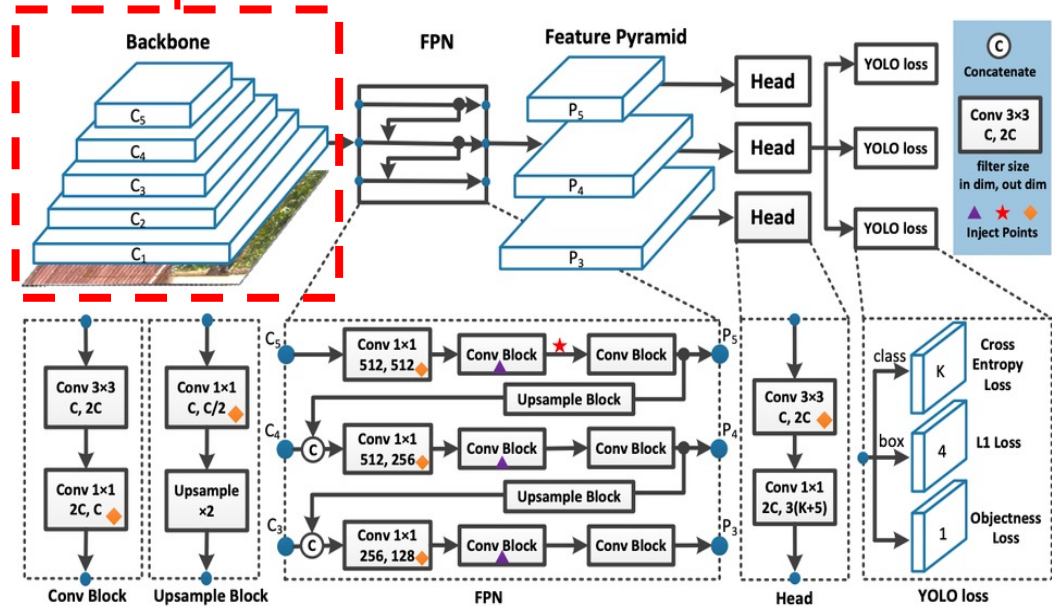
YOLOv7

3. Autonomous

◆ Real-World



Unet

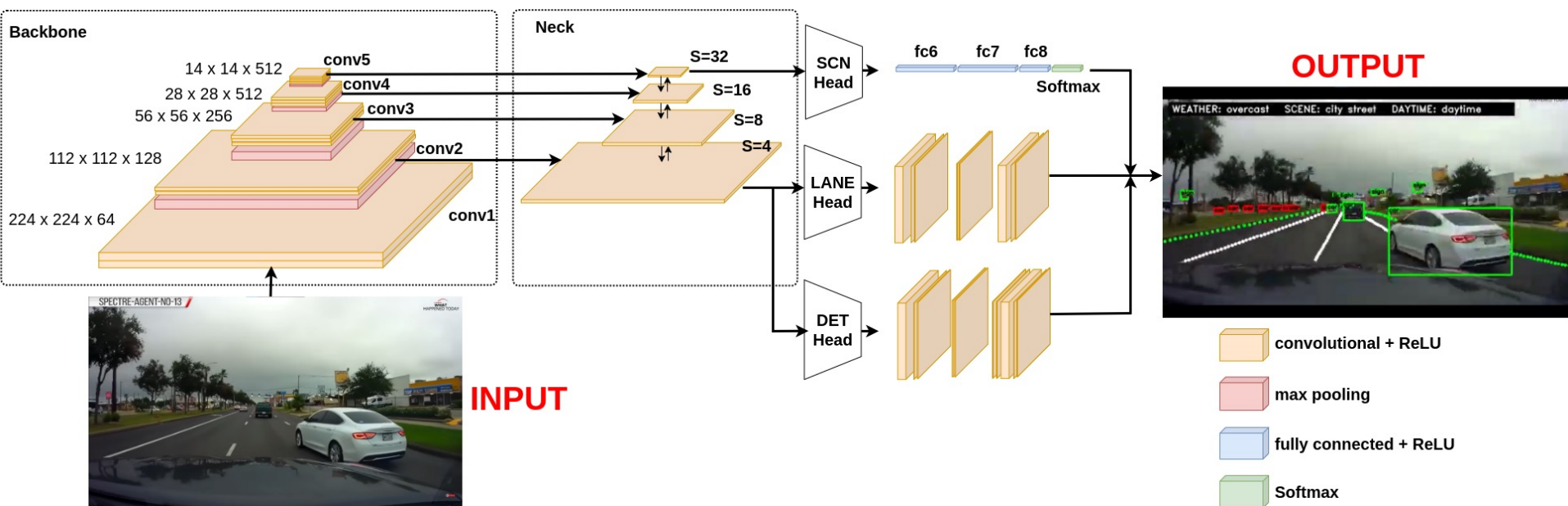


YOLOv7

3. Autonomous

◆ Real-World

Multitasking for Autonomous



Jetson Nano => **FPS: 14.63 ~ 0.068s**

Code: <https://github.com/dotrannhattuong/Multitasking-For-Autonomous>

3. Autonomous

◆ Real-World

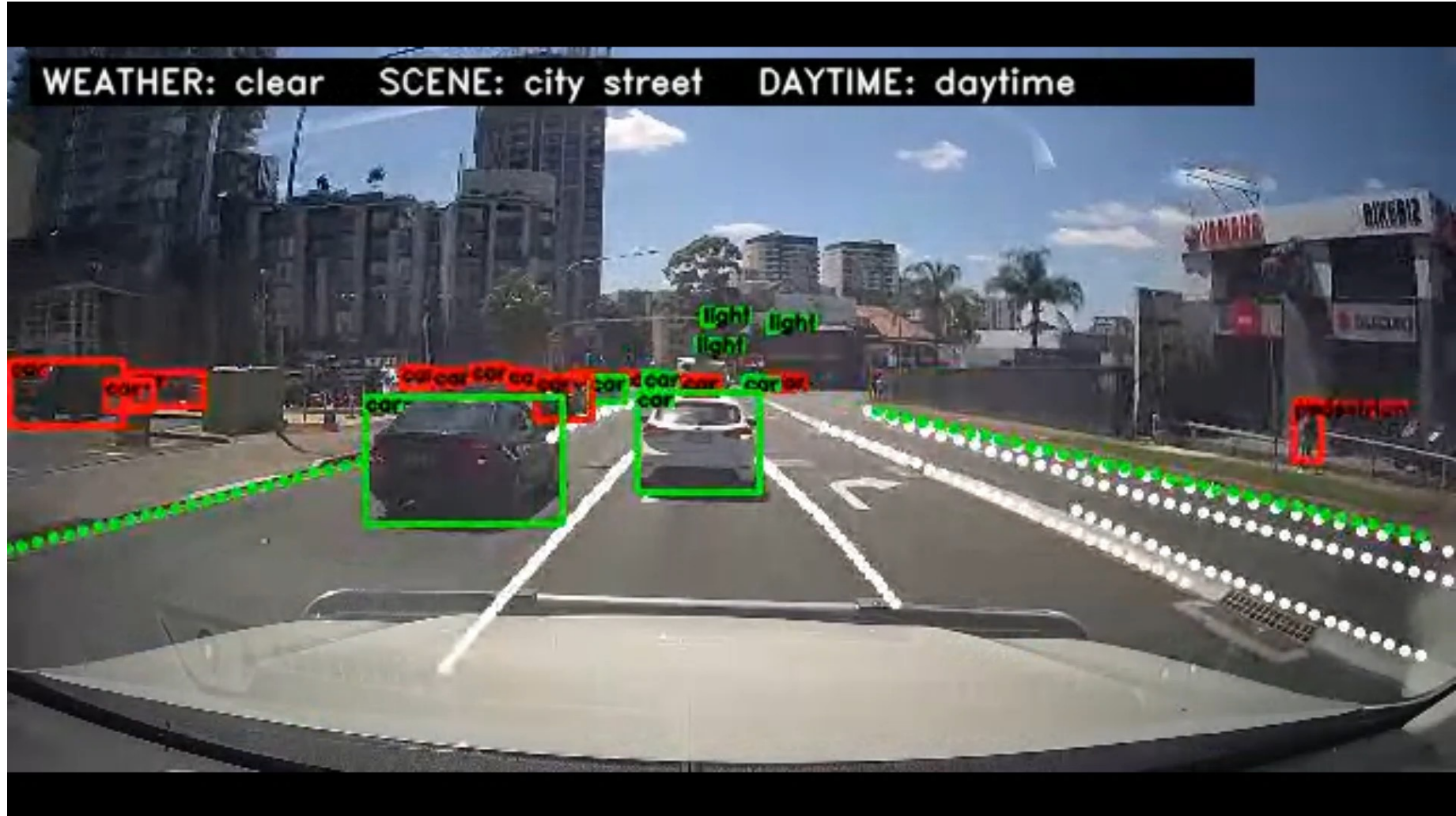
Evaluation

Mô hình	mAP (%)	IoU (%)	FPS (Jetson Nano)
YOLOv7 (Detection)	88.4	-	5.2
Unet (Segmentation)	-	94.8	-
YOLOv7 +Unet	88.4	94.8	-
Multitask	88.1	95.0	14.63

Code: <https://github.com/dotrannhattuong/Multitasking-For-Autonomous>

3. Autonomous

◆ Real-World

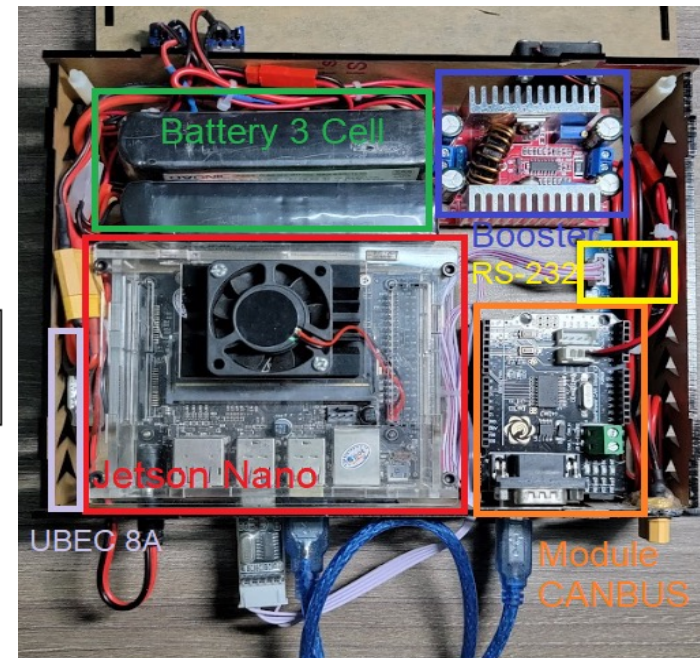
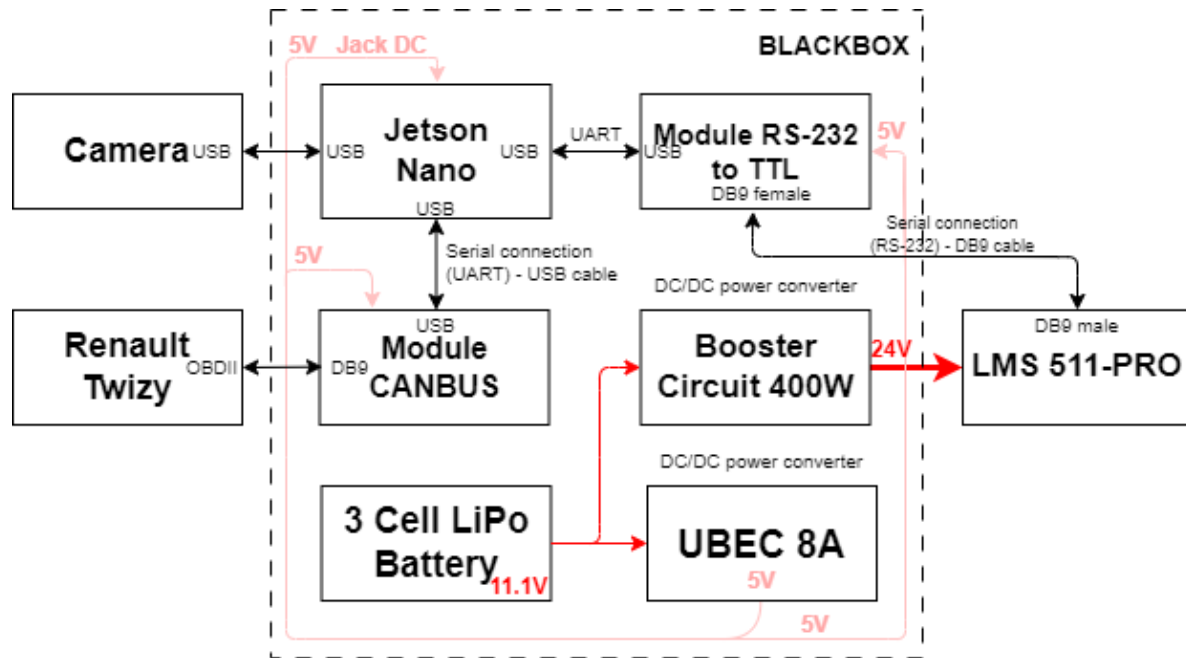


Code: <https://github.com/dotrannhattuong/Multitasking-For-Autonomous>

3. Autonomous

◆ Real-World

Hardware



Code: <https://github.com/dotrannhattuong/Multitasking-For-Autonomous>

3. Autonomous

◆ Real-World

Vehicle

- Vehicle
 - Status: DRIVING
 - Gear: D
 - Odometer: 162 Km
 - Speed: 8 Km/h
 - Battery Distance: 40 Km
 - Throttle Position: 36 %
 - Brake Pedal: INACTIVE
- Motor
 - Status: START
 - Temperature: 45 °C
 - Revolutions Per Minute: 810 RPM
 - Required Current: 11 A
- Battery
 - Charge: OFF
 - SOC: 80 %
 - Temperature: 29 °C
 - Voltage: 56 V
 - SOH: 100 %



Stop

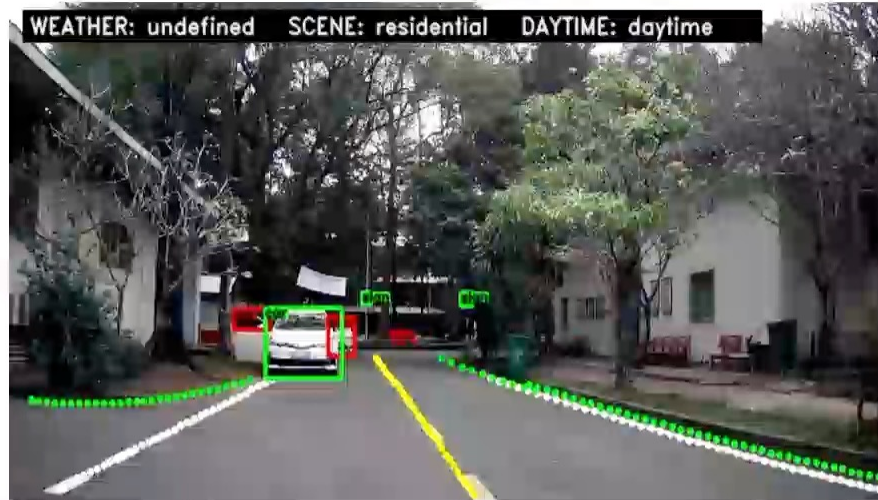
Start Client

File

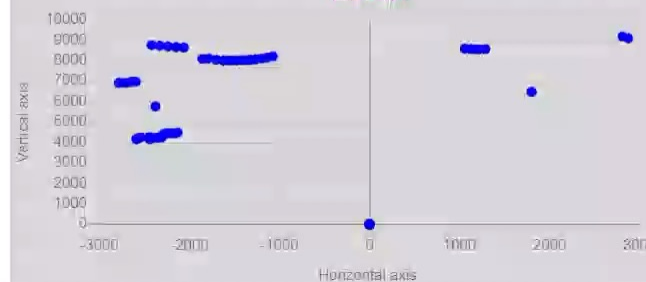
Predicting
Server

MULTITASKING FOR AUTOMONOUS

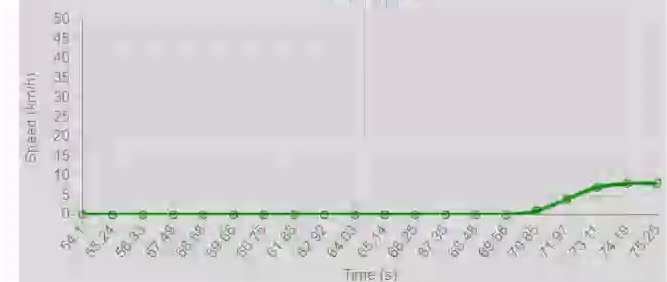
WEATHER: undefined SCENE: residential DAYTIME: daytime



LIDAR



SPEED



4. Smart Menu

◆ Problem Definition



LẨU VÀ NƯỚNGCÔ BA

005

Trương Văn Cửa TP Quy Nhơn

00778226254

THỜI

GIAN

MỞ

CỬA

13H30

23H00

CÁC

MÓN

NẤU

Lòngchiênmắm

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LòngXàonghệ

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XiahấpSả

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Xiahấpthái

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Chân gà sốt Cô Ba

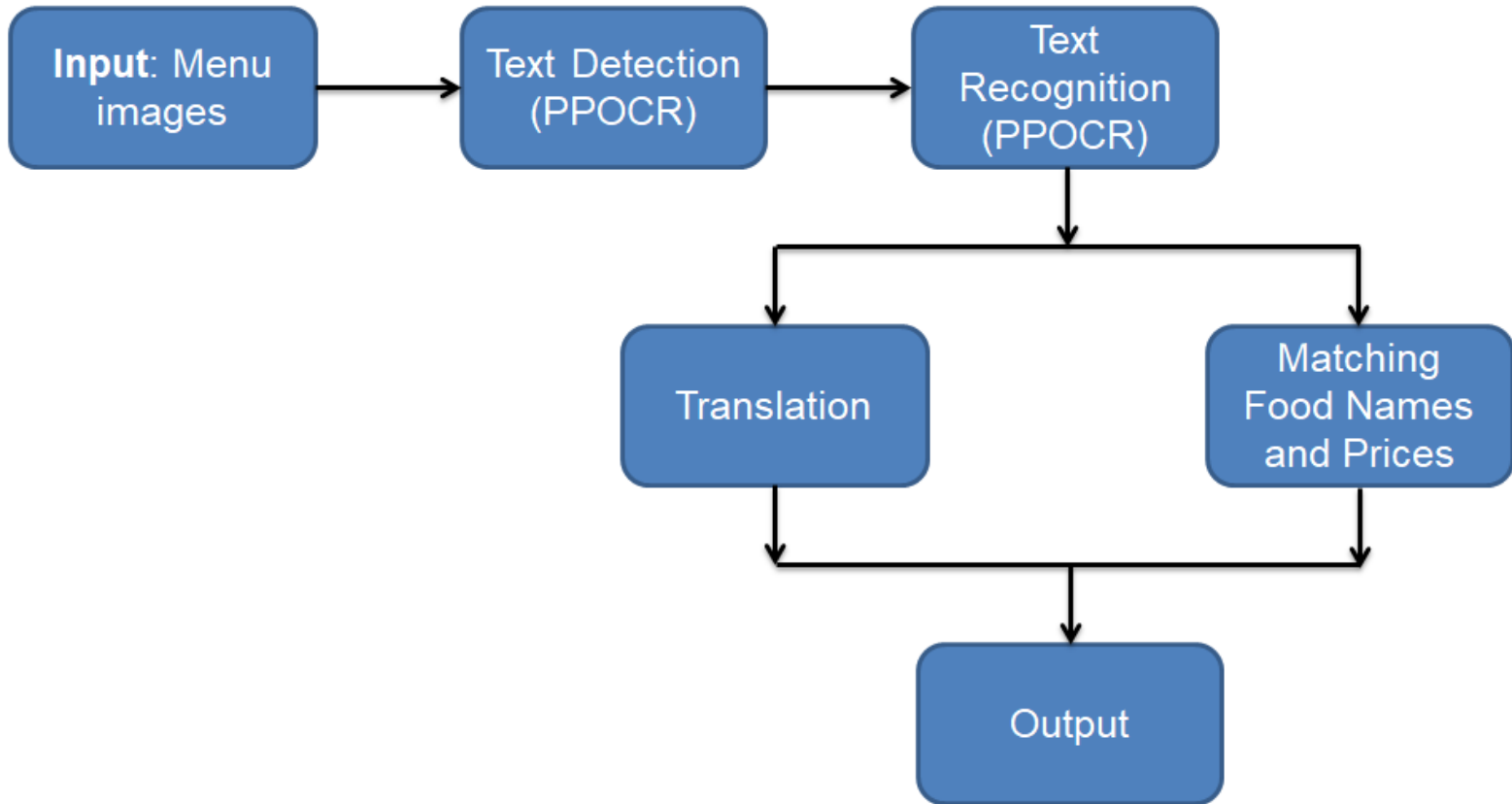
69k

Chân gà rang muối

69k

4. Smart Menu

◆ Architecture



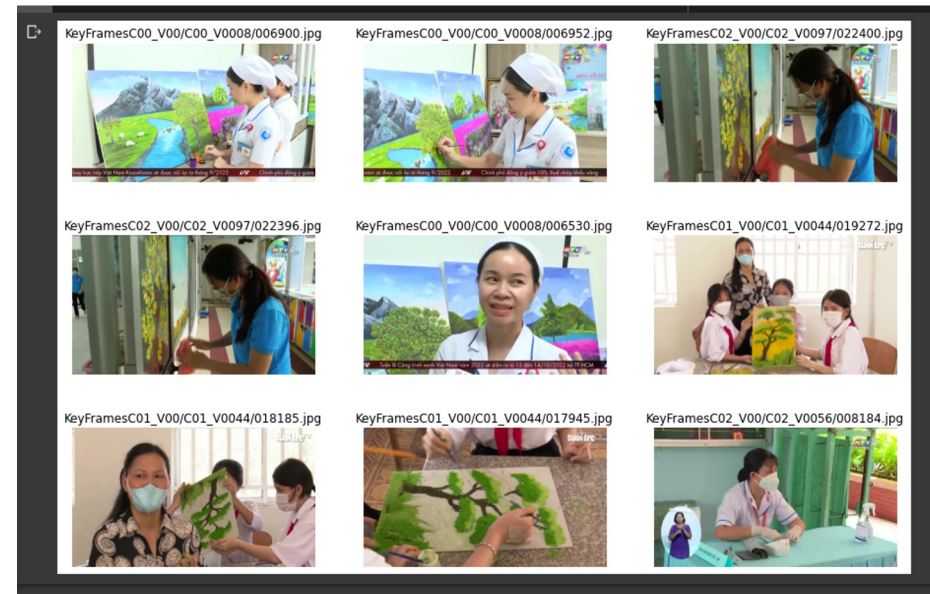
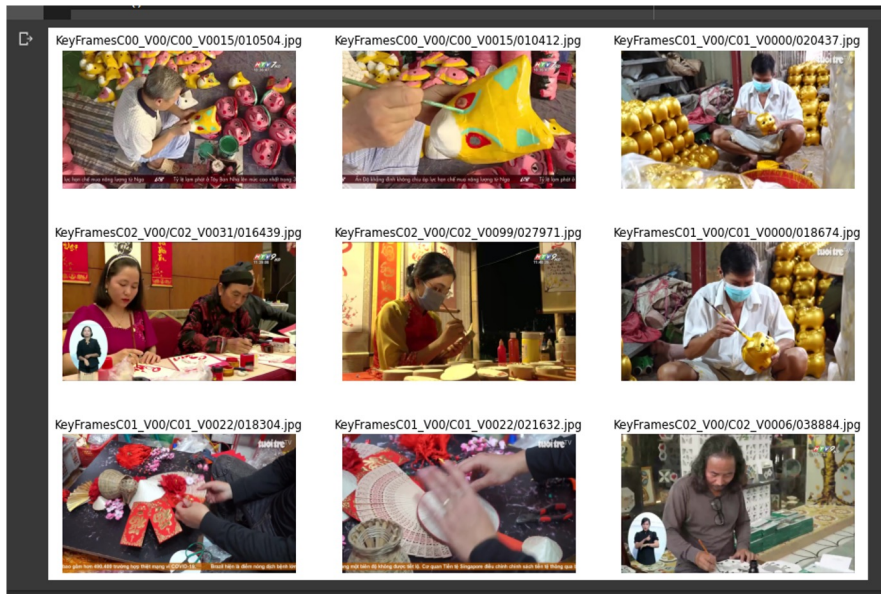
Code: https://github.com/AIVIETNAMResearch/Smart_menu_OCR

5. Retrieval System

◆ Problem Definition

Người nghệ nhân đang tô màu cho chiếc mặt nạ một cách tỉ mỉ. Xung quanh ông là rất nhiều những chiếc mặt nạ. Người nghệ nhân đi đôi dép tổ ong rất giản dị. Sau đó là hình ảnh quay cận những chiếc mặt nạ. Loại mặt nạ này được gọi là mặt nạ giấy bồi Trung thu.

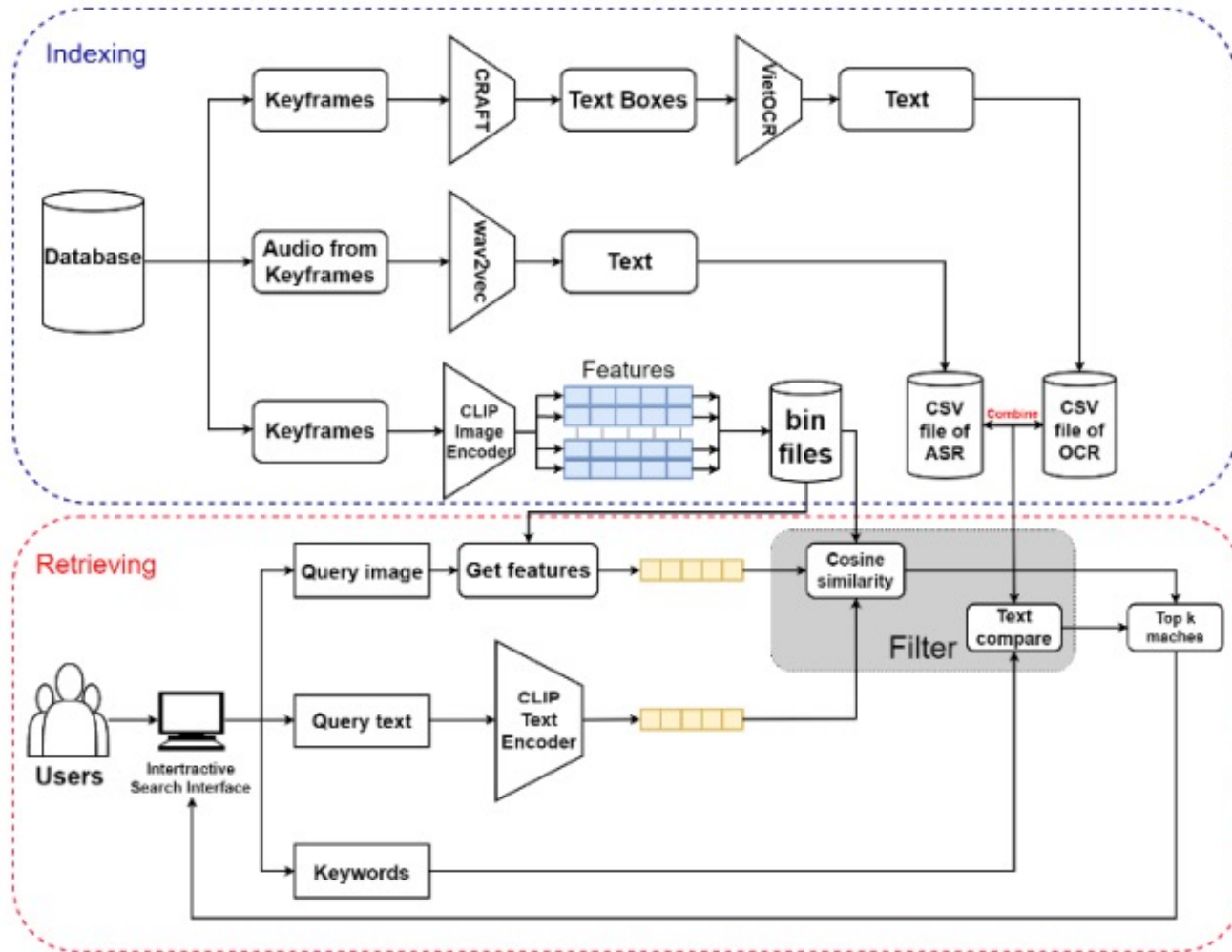
Một nữ bác sĩ/y tá đang vẽ tranh phong cảnh. Bức tranh với nền cỏ xanh có một con sông ở giữa và ngọn núi ở phía xa. Sau đó khung hình phóng gần vào bức tranh nơi đang dùng cọ để vẽ. Khung hình có hiệu ứng chuyển dần sang màu đỏ.



Code: <https://github.com/AIVIETNAMResearch/Video-Text-Retrieval>
Paper: <https://ieeexplore.ieee.org/abstract/document/10227191>

5. Retrieval System

◆ Architecture



Code: <https://github.com/AIVIETNAMResearch/Video-Text-Retrieval>

Paper: <https://ieeexplore.ieee.org/abstract/document/10227191>

5. Retrieval System

◆ Results

The screenshot displays the AIO-IR system web interface. At the top, there is a browser window with the address bar showing '127.0.0.1:5001/thumbnailimg?index=0'. Below the browser window, there is a search bar with the placeholder text 'Text query' and buttons for 'Search', 'Clear', and 'Download'. Below the search bar, there is a section for image path and search parameters. The 'IMAGE PATH' field contains 'C00_V0130/028539'. To its right is a 'Search' button. Further right is the 'SEARCH ASR' field, which is empty, followed by another 'Search' button. To the right of these is the 'WRITE CSV' section with a dropdown menu set to 'Single'. To the right of that is the 'NUM OF QUERY' section with a dropdown menu set to '1' and a 'Visualize' button. Below these fields, there is a grid of 12 image thumbnails. Each thumbnail is labeled with a file path, such as 'C00_V0000/000000.jpg', 'C00_V0000/000039.jpg', 'C00_V0000/000079.jpg', 'C00_V0000/000096.jpg', 'C00_V0000/000118.jpg', 'C00_V0000/000158.jpg', 'C00_V0000/000160.jpg', 'C00_V0000/000170.jpg', 'C00_V0000/000181.jpg', 'C00_V0000/000191.jpg', 'C00_V0000/000202.jpg', and 'C00_V0000/000204.jpg'. The thumbnails show various images, including a person's face, a crowd, and a logo with the text 'giây'.

6. Research

◆ Domain Adaptation - CVPR 2024 ~ 6 months

- ViT branch includes a ViT encoder $E_1(\cdot, \theta_{E_1})$ and a classifier $F_1(\cdot, \theta_{F_1})$
- CNN branch includes a CNN encoder $E_2(\cdot, \theta_{E_2})$ and a classifier $F_2(\cdot, \theta_{F_2})$

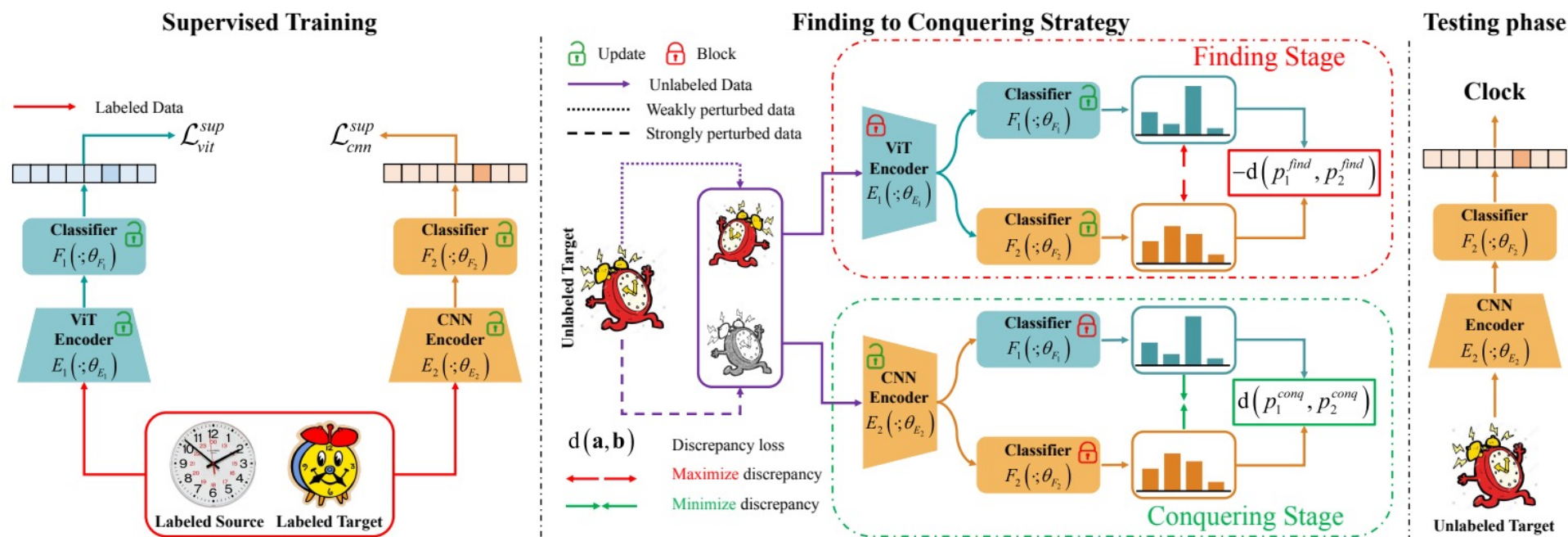


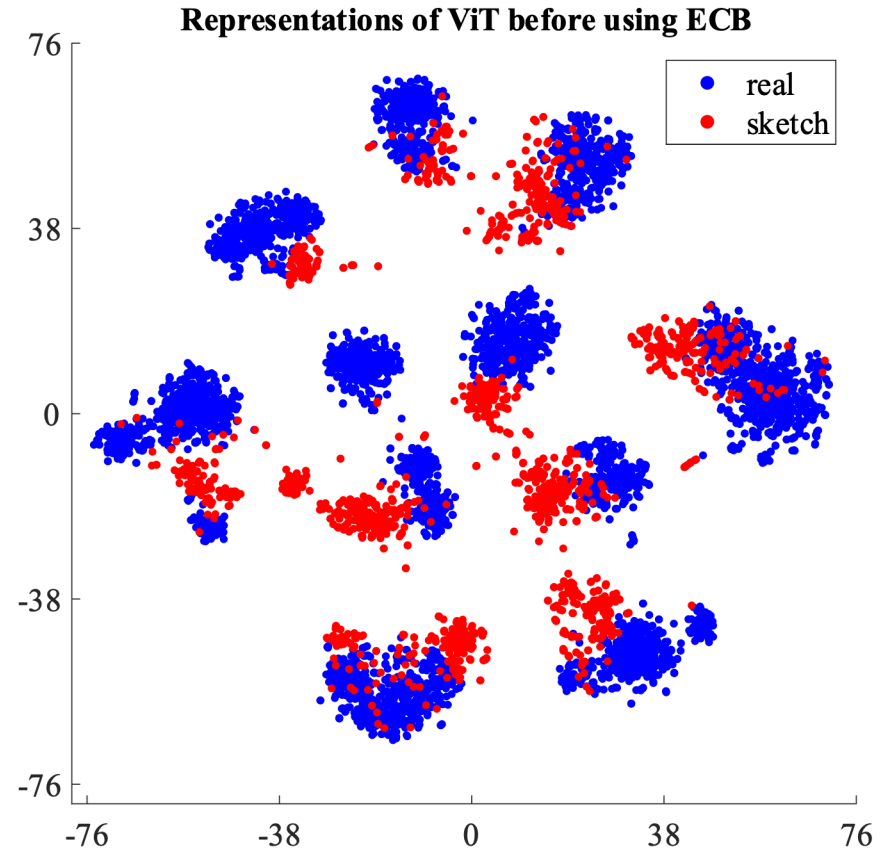
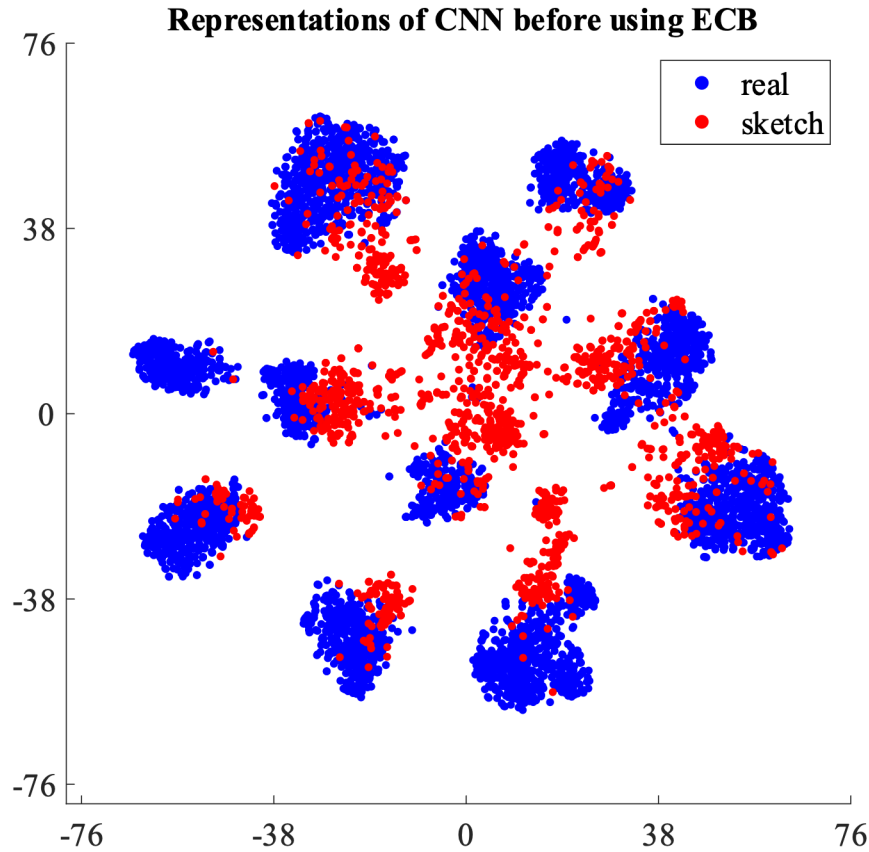
Figure 2. Illustration of a hybrid network with the proposed Finding to Conquering strategy. We use ViT to build E_1 that drives two classifiers F_1 and F_2 to expand class-specific boundaries comprehensively. Besides, we select CNN for the second encoder E_2 to cluster target features based on the boundaries identified by ViT. These encoders all use two classifiers F_1, F_2 .

Project page: <https://dotrannhattuong.github.io/ECB/website/>

Paper: https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_paper.html

6. Research

◆ Domain Adaptation: Visualization t-SNE

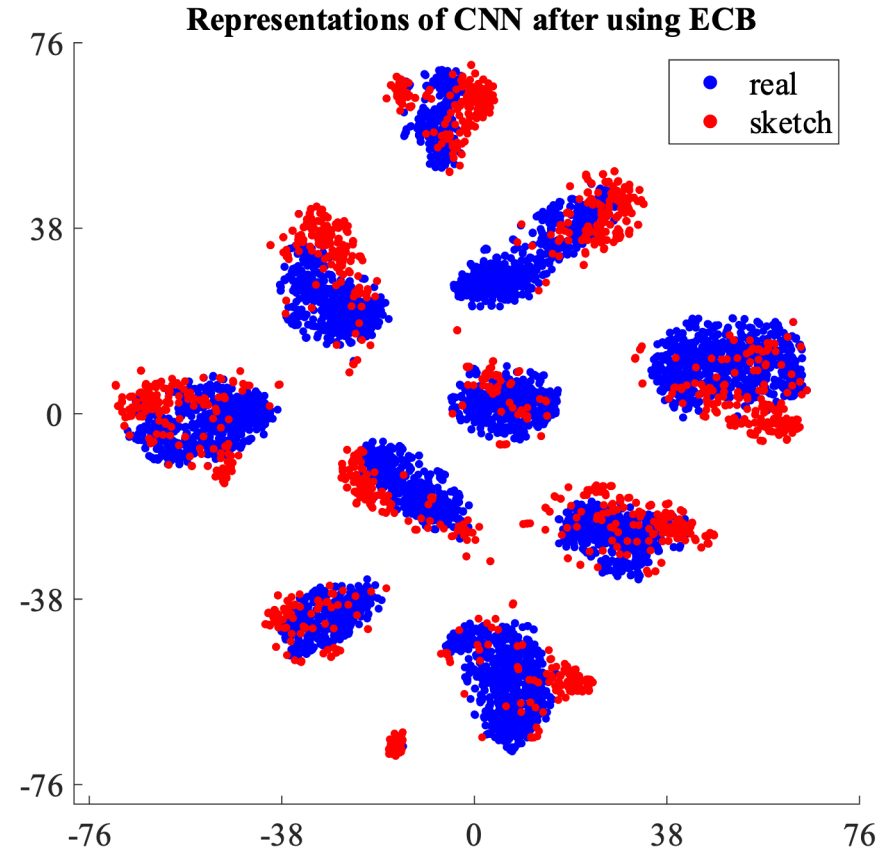
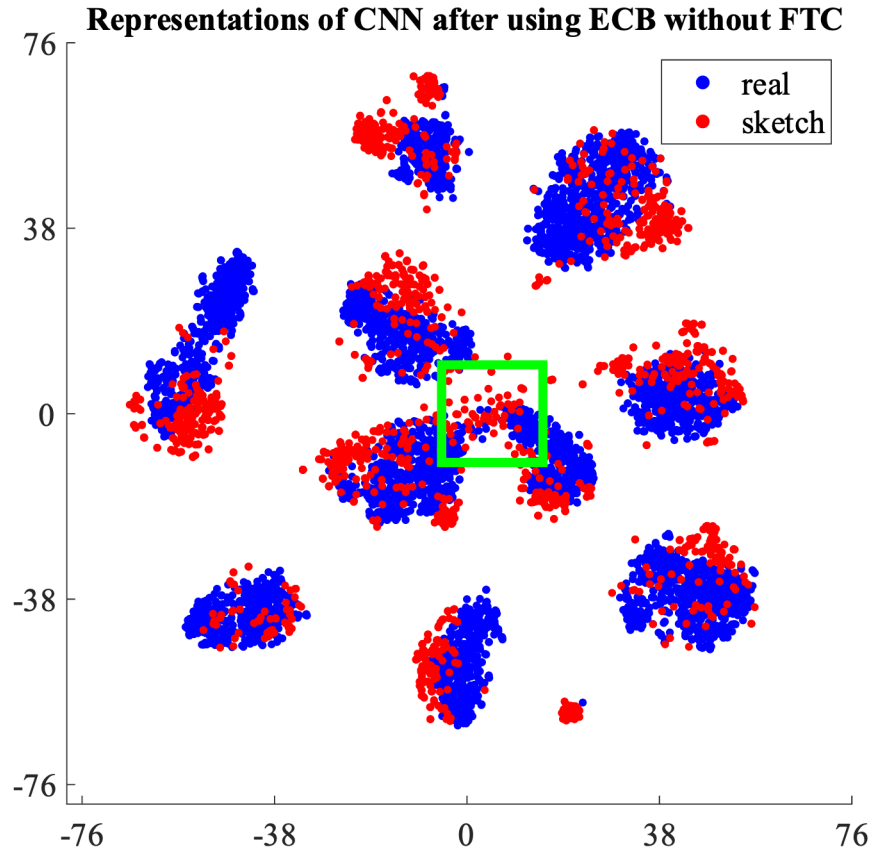


Project page: <https://dotrannhattuong.github.io/ECB/website/>

Paper: [https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_p
aper.html](https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_paper.html)

6. Research

◆ Domain Adaptation: Visualization t-SNE



Project page: <https://dotrannhattuong.github.io/ECB/website/>

Paper: [https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_p
aper.html](https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_paper.html)

6. Research

◆ Domain Adaptation: Attention map visualization

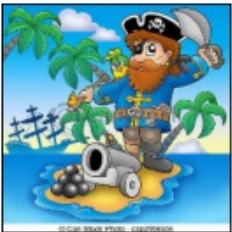
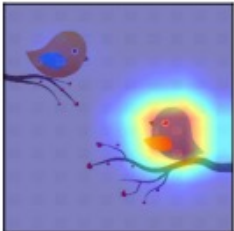
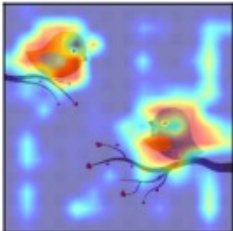
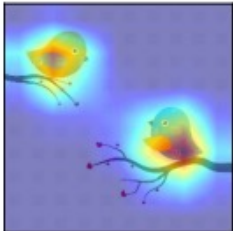
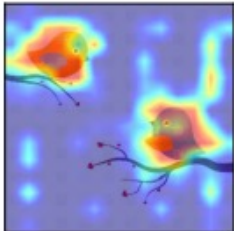
Cannon	CNN (Before)	ViT (Before)	CNN (After)	ViT (After)
				
Bird	CNN (Before)	ViT (Before)	CNN (After)	ViT (After)
				

Table 4. **Visualize the feature maps** for the ‘*Cannon*’ and ‘*Bird*’ examples to investigate the learning behaviors of CNN and ViT with and without using the proposed method ECB.

Project page: <https://dotrannhattuong.github.io/ECB/website/>

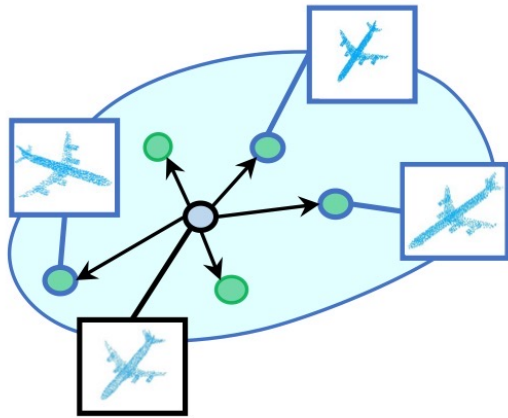
Paper: [https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_p](https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_paper.html)
[aper.html](https://openaccess.thecvf.com/content/CVPR2024/html/Ngo_Learning_CNN_on_ViT_A_Hybrid_Model_to_Explicitly_Class-specific_CVPR_2024_p)

6. Research

◆ Data Augmentation on Point Cloud

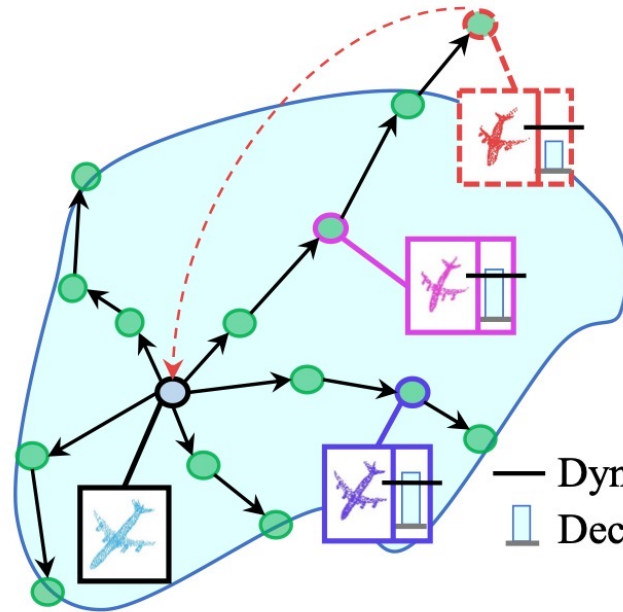
Motivation

Previous Methods



- Original sample
- Unreliable sample
- Data Augmentation
- Class-decision boundary
- Back to original sample
- Augmented samples

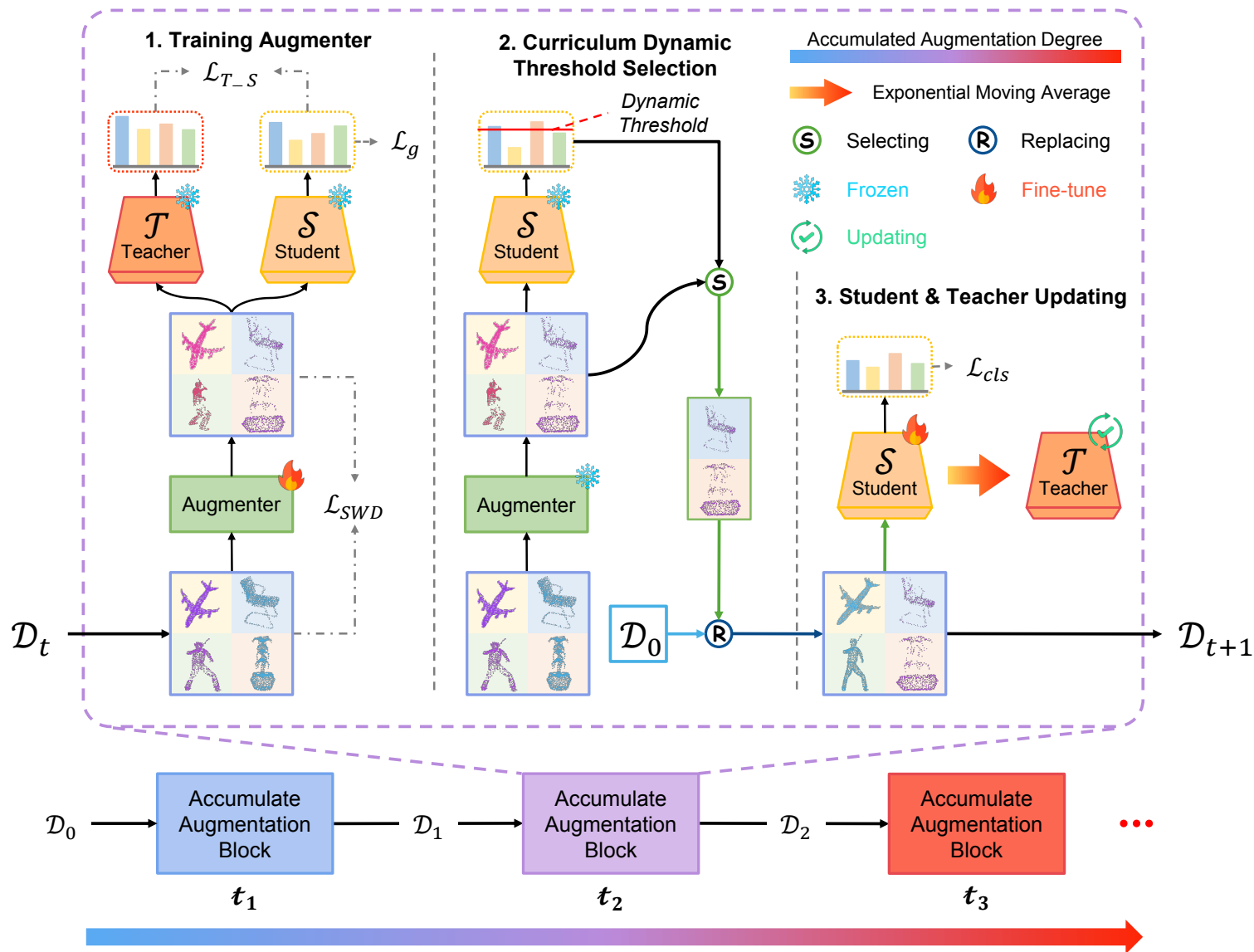
DA^2



- Dynamic threshold
- Decision probability

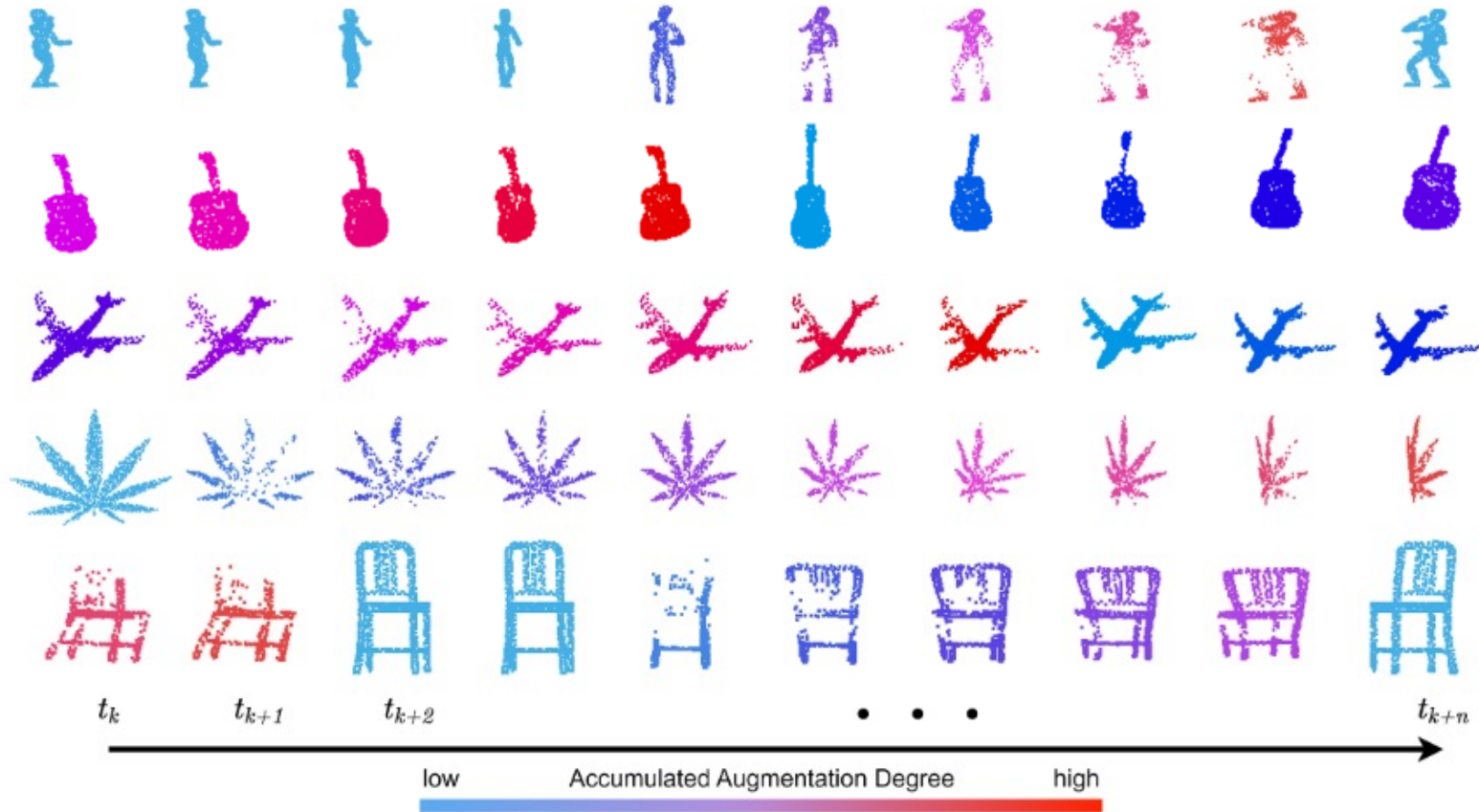
6. Research

◆ Data Augmentation on Point Cloud



6. Research

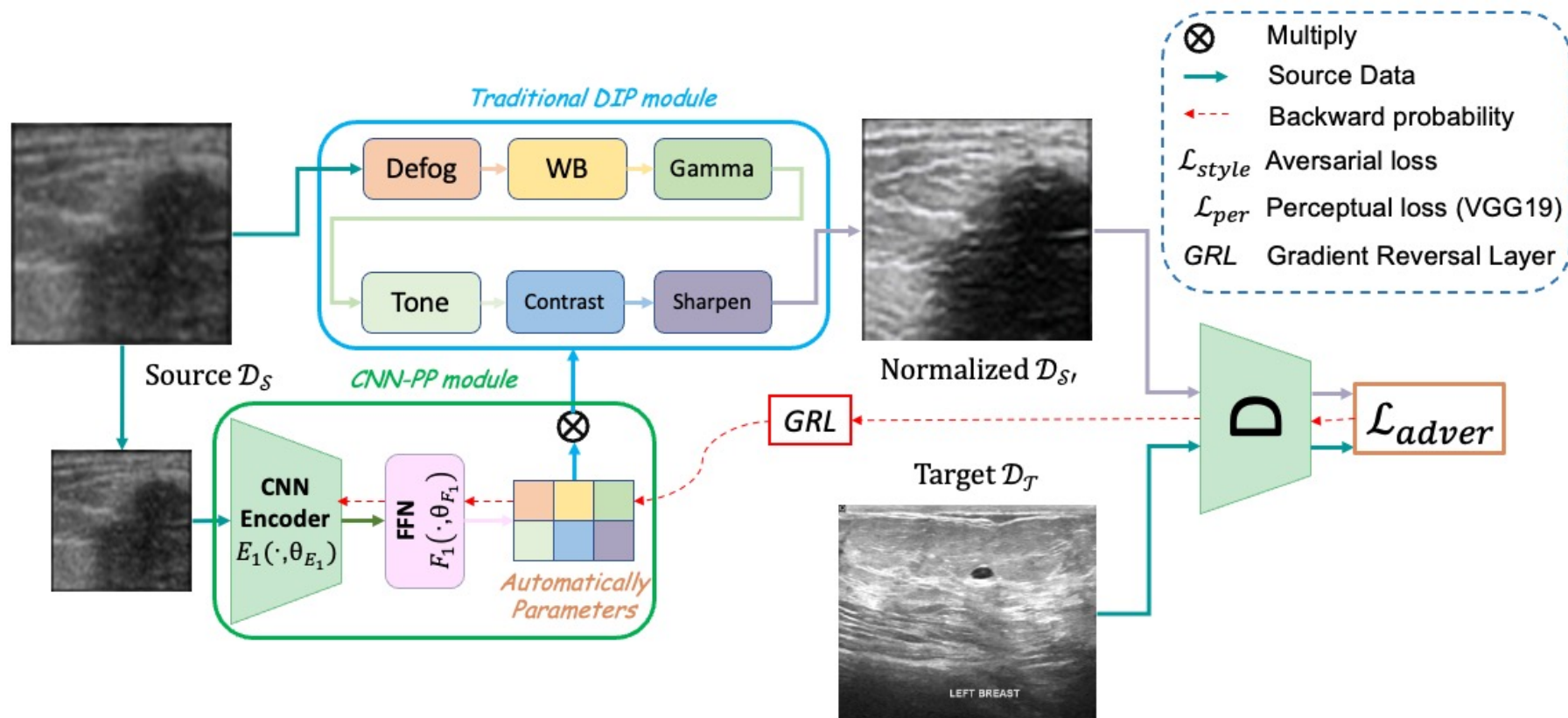
◆ Data Augmentation on Point Cloud



6. Research

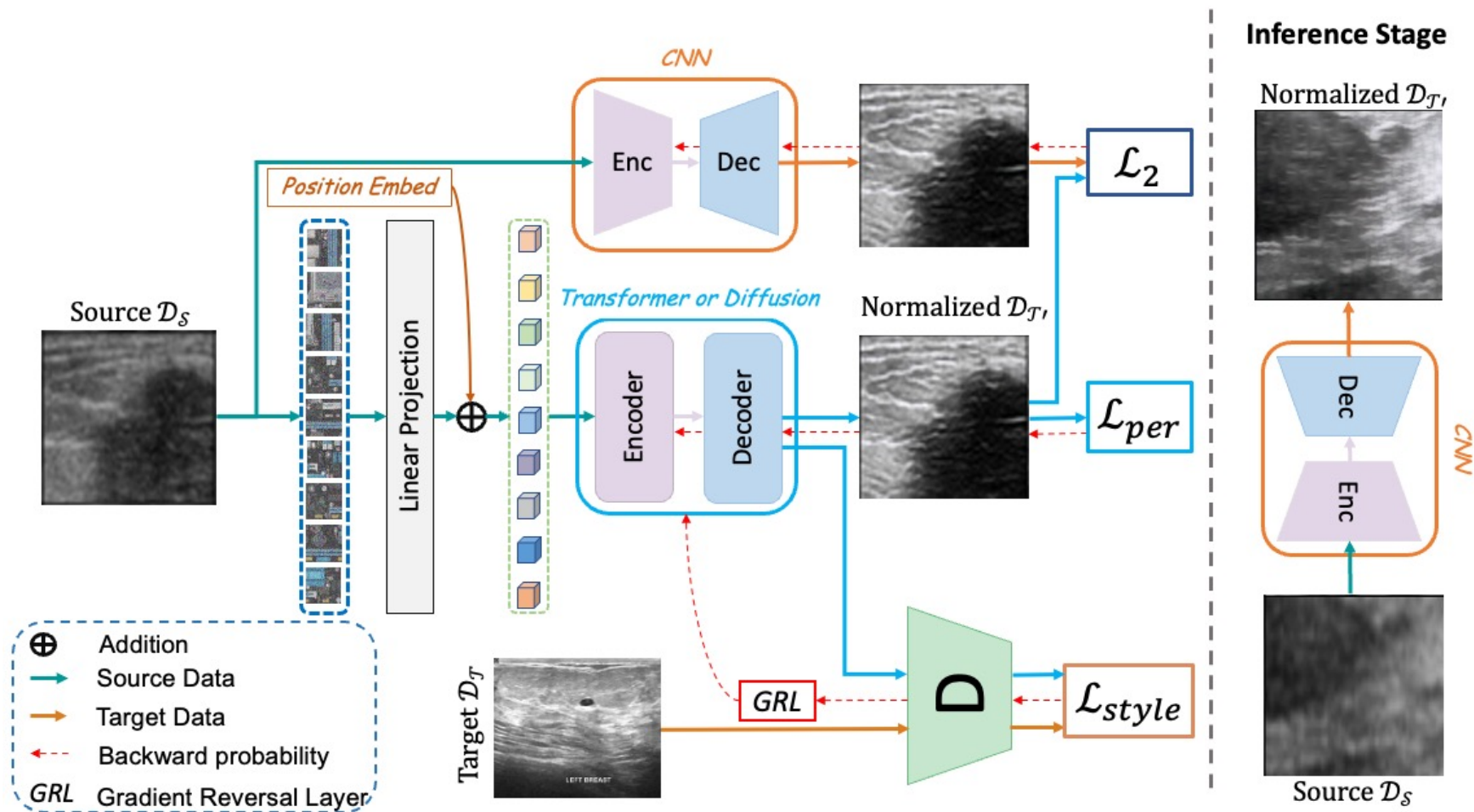
◆ Style Transfer

Using DIP module



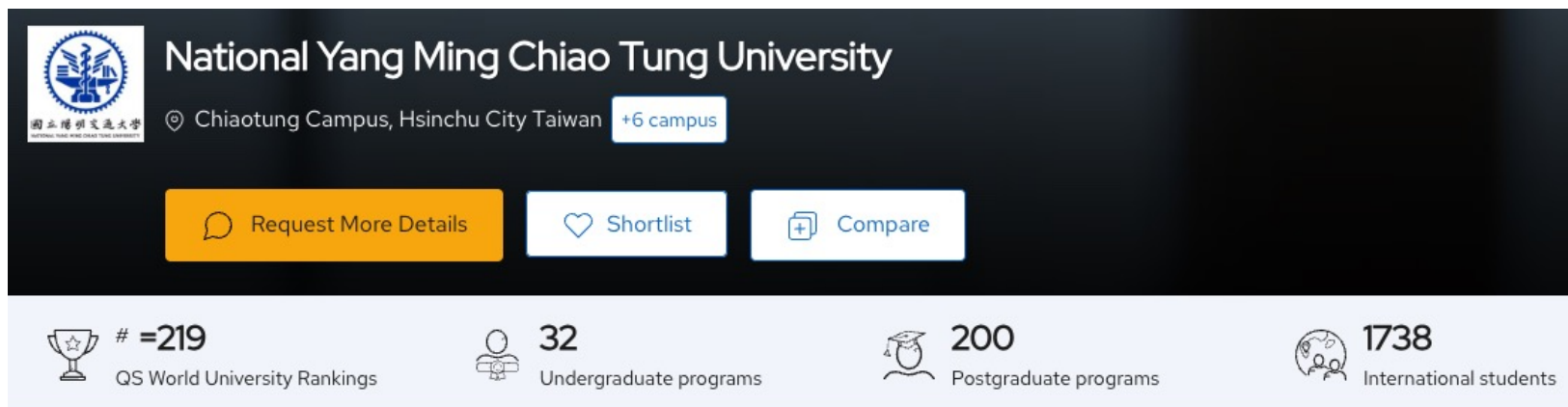
6. Research

◆ Style Transfer



7. Education

◆ Master Student at *National Yang Ming Chiao Tung University*



The screenshot shows the university's profile on a platform. It includes the university's logo, name, and location (Chiaotung Campus, Hsinchu City Taiwan). Below this are three buttons: 'Request More Details', 'Shortlist', and 'Compare'. At the bottom, there are four statistics: QS World University Rankings (#219), Undergraduate programs (32), Postgraduate programs (200), and International students (1738).

Ranking	Undergraduate programs	Postgraduate programs	International students
#219	32	200	1738
QS World University Rankings			

Application Documents

- Application form (Both the applicant and the home university coordinator need to [sign on the last page.](#))
- Passport information page (valid for at least 3 months)
- Curriculum Vitae (CV) in English
- Study Plan ([sample](#))
- Official Transcripts (in English or Chinese)
- Latest enrollment letter from the home institution or a student card with an expiry date
- Recommendation Letter from the home institution (optional)
- Other supporting documents (optional)

Ranking: <https://www.topuniversities.com/universities/national-yang-ming-chiao-tung-university>

Admission: <https://oia-sys.nycu.edu.tw/admission/>

Requirements: <https://oia.nycu.edu.tw/oia/en/app/artwebsite/view?module=artwebsite&id=821&serno=a277c341-8d25-4a51-8495-58d1d58bbc5d>

7. Education

◆ Advisor



黃敬群
Ching-Chun Huang
Professor

About

- Education - Ph.D., M.S. and B.S. in EE, National Chiao Tung University, Taiwan
- <http://acm.cs.nycu.edu.tw/>
- chingchun[at]cs.nycu.edu.tw
- EC708
- 03-5712121 # 54736

Research Interest

Machine Learning/Deep learning, Autonomous Vehicle, Medical Image Processing, Intelligent Building, Augmented reality/ Virtual reality

Lab

- Applied Computing and Multimedia Lab
- EC118
- 03-5712121 # 54769
- <http://acm.cs.nycu.edu.tw/>

Prof. Ching-Chun Huang: <https://www.cs.nycu.edu.tw/members/detail/chingchun?locale=en>

Lab: <http://acm.cs.nycu.edu.tw/>

7. Education

◆ Research Topics in the lab:

1. Large language model (LLM).
2. 3D Generation.
3. Self-Driving Car.
4. Generative Model.

7. Education



Thank you for listening